Assignment 6

Collateralised FX and interest-rate products

Aditya Amit Shah | as22008
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1 / Valuation formula

Let t be today, T the settlement date, and E, U, G denote EUR, USD, GBP. Let S_t be spot USD per EUR and X_t spot USD per GBP. The position receives N EUR and pays NK USD at T, where K is USD per EUR. The opposite position reverses the sign.

$P^{i|k}(t,T)$ is the value **in currency i** of one unit of currency i paid at T under collateral currency k. Assume full, symmetric, continuously adjusted GBP cash collateral remunerated at SONIA, zero threshold/minimum transfer amount, and no default, collateral optionality, or transaction costs.

Value the two terminal currency payments separately under GBP collateral, convert the EUR receipt into USD at current spot, and subtract the USD payment:

$$V_U^G(t) = N\left[S_t P^{E\,|\,G}(t,T) - K P^{U\,|\,G}(t,T)\right]$$

Equivalently, convert the terminal USD payoff into GBP before taking the collateral-discounted expectation:

$$V_U^G(t) = N X_t \,\mathbb{E}_t^{Q_G}\left[e^{-\int_t^T c_G(u)\,du}\frac{S_T - K}{X_T}\right]$$

Here Q_G is the currency-GBP risk-neutral pricing measure and c_G is SONIA collateral remuneration. Discounting a USD payoff directly at SONIA omits the terminal FX conversion. GBP reporting gives $V_G{}^G = V_U{}^G / X_t$.

2 / Fair forward strike

Setting the inception value to zero gives the fair strike and the equivalent mark-to-market expression:

$$K_G^*(t,T) = S_t \frac{P^{E\,|\,G}(t,T)}{P^{U\,|\,G}(t,T)}$$

$$V_U^G(t) = N P^{U\,|\,G}(t,T)\left[K_G^*(t,T) - K\right]$$

A contracted strike below the fair strike gives positive PV to the receive-EUR position. At maturity, both discount factors equal one and the value becomes $N(S_T - K)$. These formulas describe a single terminal exchange.

3 / Sufficiency of the five supplied curves

The curves are sufficient in a consistent deterministic collateral-basis framework, given spot FX. Without a collateral-switch assumption, they are not sufficient in general.

Full discount-factor convention

The PDF does not define the bracket ordering. First take [i,U] to mean cashflow currency i collateralised in USD, and interpret the bracket quantity as a full discount factor. Define all factors at the same (t,T):

Symbol	Supplied discount factor
D_G	SONIA (GBP under GBP collateral)
D_E	EuroSTR (EUR under EUR collateral)
D_U	SOFR (USD under USD collateral)
A_G	$df^{[GBP,USD]}$ (GBP under USD collateral)
A_E	$df^{[EUR,USD]}$ (EUR under USD collateral)

Derive the collateral switch

Write $y_i = r_i - c_i$, where r_i is the formal currency-i financing/risk-free rate and c_i is its collateral remuneration. A currency-i claim under currency-k collateral has effective discount rate $r_i - y_k$.

Assume $b_{U,G} = y_U - y_G$ is deterministic over [t,T]. Switching from USD collateral to GBP adds this same spread for every payout currency. Its exponential factors out of each pricing expectation:

$$P^{i\,|\,G} = L_{G\leftarrow U} P^{i\,|\,U}, \qquad L_{G\leftarrow U} = e^{-\int_t^T (y_U - y_G)\,du}$$

Apply the identity to a GBP payment to obtain $L = D_G/A_G$. Then recover both required GBP-collateral curves:

$$L_{G\leftarrow U} = \frac{D_G}{A_G}, \qquad P^{U\,|\,G} = \frac{D_G D_U}{A_G}, \qquad P^{E\,|\,G} = \frac{D_G A_E}{A_G}$$

Substitution into Questions 1 and 2 gives:

$$K_G^* = S_t \frac{A_E}{D_U}, \qquad V_U^G = N \frac{D_G}{A_G}(S_t A_E - K D_U)$$

The switch factor cancels from the fair strike but scales an off-market value. EuroSTR is redundant here because A_E already incorporates EUR/USD collateral basis; A_E/D_E extracts that adjustment. SONIA alone cannot discount both currency legs.

3 continued / Notation and model limits

If the bracket quantities are basis-only factors

Some curve libraries use $df^{[i,U]}$ for a multiplicative basis adjustment rather than a full currency discount factor. Define $B_{i,U} = P^{i|U}/D_i$. Convert to full curves first:

$$B_{i,U} = \frac{P^{i\,|\,U}}{D_i}, \qquad A_G = D_G B_{G,U}, \qquad A_E = D_E B_{E,U}$$

The deterministic switch construction is then:

$$P^{U\,|\,G} = \frac{D_U}{B_{G,U}}, \qquad P^{E\,|\,G} = \frac{D_E B_{E,U}}{B_{G,U}}$$

$$K_G^* = S_t \frac{D_E B_{E,U}}{D_U}, \qquad V_U^G = \frac{N}{B_{G,U}}(S_t D_E B_{E,U} - K D_U)$$

These are the same formulas expressed under the alternative input convention. A basis-only factor must not be substituted for a full discount factor. If the bracket ordering is reversed, the inputs must likewise be reinterpreted before use.

Why the unrestricted stochastic answer differs

In the general foreign-collateral pricing relation,

$$P^{i\,|\,k}(t,T) = \mathbb{E}_t^{Q_i}\left[e^{-\int_t^T (r_i(u) - y_k(u))\,du}\right]$$

a stochastic collateral-switch exponential stays inside the expectation. The ratio D_G/A_G does not determine the corresponding switch expectation for each other payout currency. The five marginal curves therefore do not identify the required GBP-collateralised EUR and USD curves without an additional joint FX/rate/basis model or directly calibrated GBP-collateral curves.

Forward triangulation uses a common collateral currency; it does not, by itself, permit switching collateral. Thus the strike cancellation on page 2 is a result of the deterministic switch assumption, not a universal no-arbitrage identity. The individual interest rates need not all be deterministic.

Illustrative one-year check

A separate synthetic one-year example in the notebook checks the two input representations, zero value at the fair strike, and the PV sign. With spot 1.10 USD per EUR and EUR notional 1,000,000, the illustrative curves produce a fair strike of 1.118859855 and USD PV 8,529.496637 at K = 1.11. These are separate from the supplied swap data.

Reference: Fujii and Takahashi (2015), A General Framework for the Benchmark Pricing in a Fully Collateralized Market, Theorem 2.1 and Section 5.1. arxiv.org/abs/1508.06339. The deterministic conversion is derived above from the stated spread assumption.

4 / Completed implementation and results

All six assigned methods are complete. The supplied FRA example, stream engines, public interfaces, inputs, and registrations are retained. Overnight coupons use their index projection curves; both legs discount on the funding curve.

Library file	Completed methods
product/linear_products.py	ProductRFRSwap.__init__: fixed and overnight legs. ProductOvernightIndexBasisSwap.__init__: both overnight legs; spread only on leg 1.
product/product_factory.py	create_rfr_swap and create_overnight_index_basis_swap: convention fields, quote, dates, notional and direction.
yield_curve/valuation_engine.py	Both swap calculate_value methods: signed PV and cash aggregation, with totals reset on each calculation.

Default results (USD)

Value date: January 2, 2026. Notional: 1,000,000 USD. Accrual start: April 2, 2026. Swap accrual end: April 2, 2027. Fixed rate: 4%; basis spread over SOFR: 10 bp. The FRA accrual ends July 2, 2026.

Product / leg	PV (USD)
FRA: receive fixed	87.090294207
RFR fixed leg	38,534.539463150
RFR floating leg	41,164.219037938
RFR swap: receive fixed	-2,629.679574788
Basis leg 1: SOFR + spread	42,163.002863421
Basis leg 2: Fed Funds	33,716.530392254
Basis swap: receive leg 1	8,446.472471167

Validation

All **18 notebook code cells** executed in order from a fresh kernel, including the supplied formula and factory-equivalence assertions. All **13 independent tests** passed. Tests cover analytic PVs on distinct projection/funding curves; fixed versus floating day counts; spread placement; pay/receive signs; cash today and expiry; payment lag; repeated valuation; frequencies; factory date and convention mappings; serialization; and signed cashflow reports.

Validated with Python 3.13.6, NumPy 2.3.2, pandas 2.3.1, and QuantLib 1.41. The starter recommends Python 3.11 and its pinned requirements; the local run used compatible installed Python 3.13 packages.

The complete executed notebook, readable HTML, library source, and tests are included together. This report summarises the written answers and implementation results; retain the full project for a code submission.